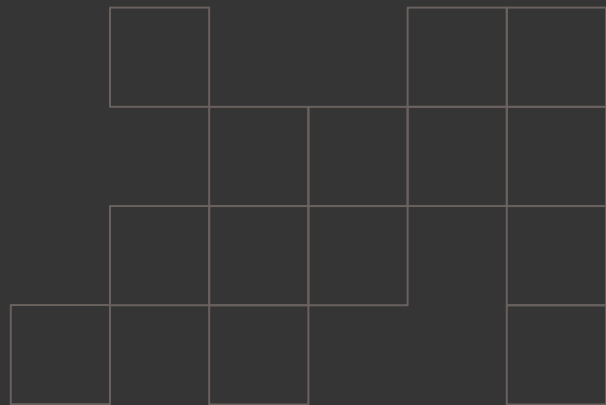


Investigating the Relationship Between Playing Style and Expected Goals

By: Christian Haranzo, Teresa Bui, Rajeswari Naishadham, Michael Le

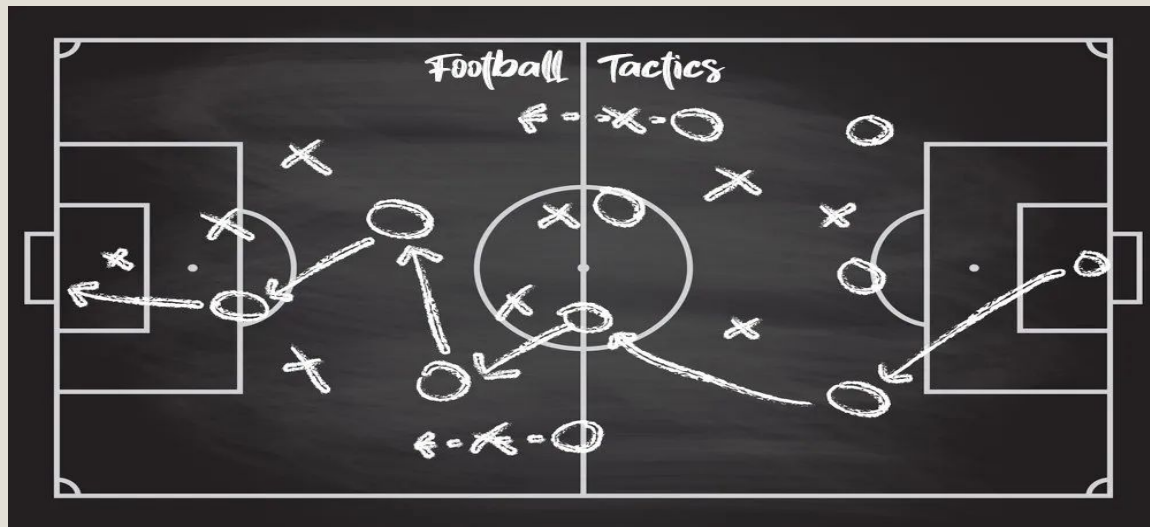


Introduction

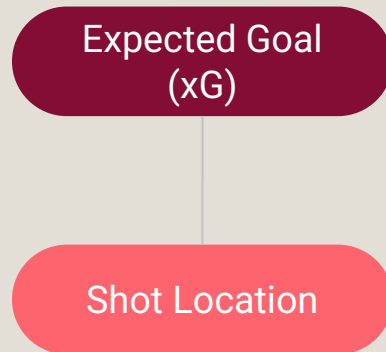
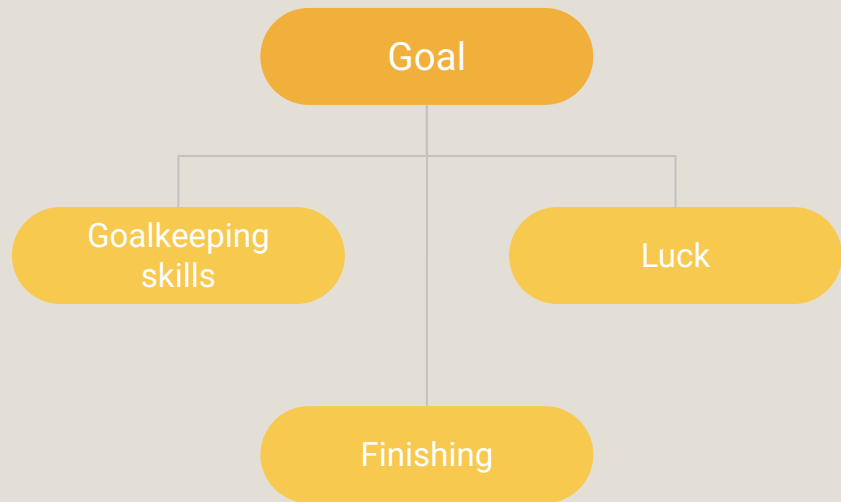
Which tactical style creates the highest-quality scoring chances?

Research Question: To what extent are the four tactical concepts: possession-based build-up, attacking width, vertical attacking movements, and defensive aggression, playing style indicators associated with a team's expected goals (xG) during the FIFA World Cup 2026?

Goal: Identify what methods are highly associated with higher expected goals (xG) that can help team leaders make better decisions during trainings and live matches.



Why xG, not goals?

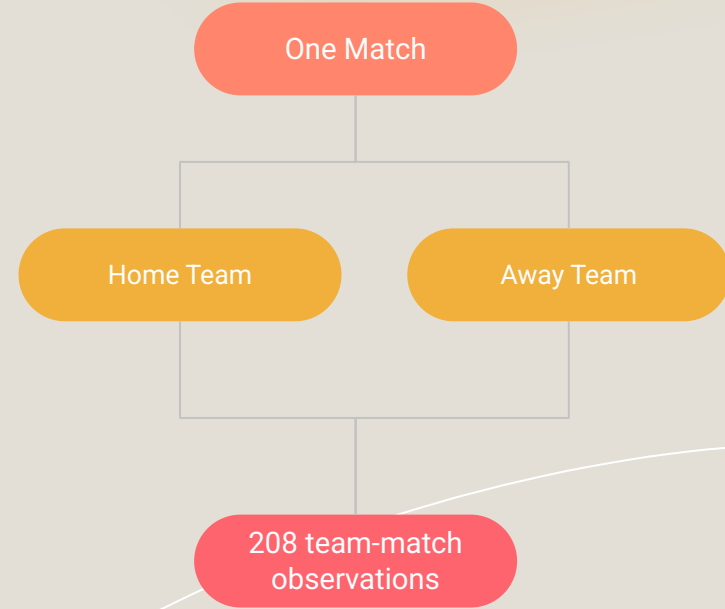


xG measures chance *quality*, isolating it from finishing luck.

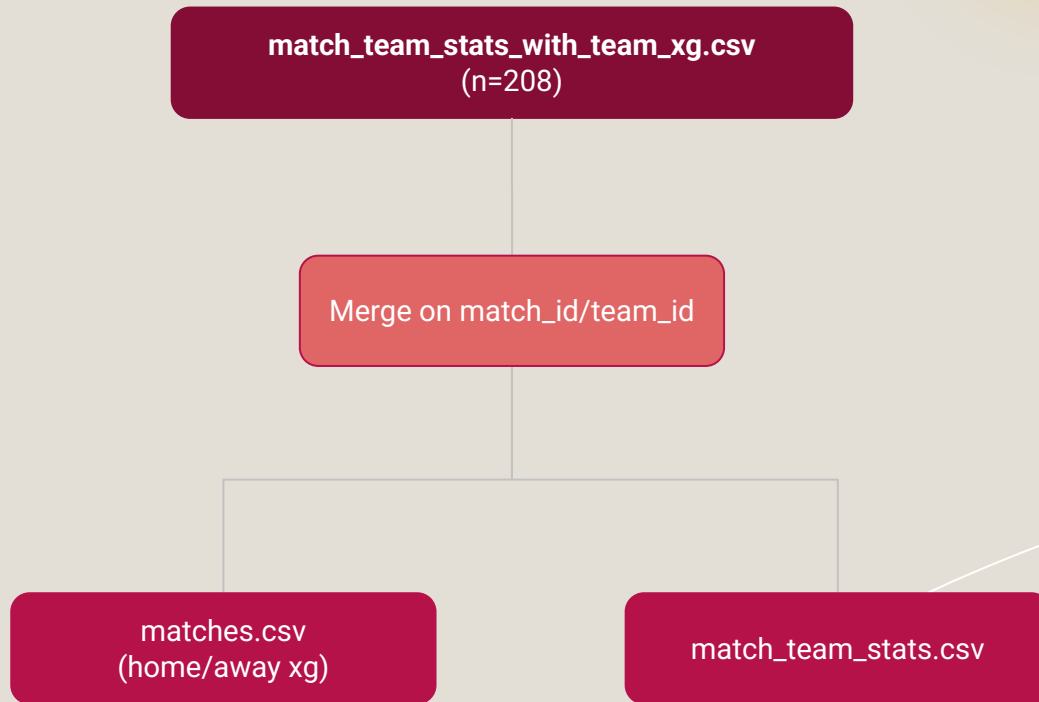
Four Tactical Concepts

Tactical Concept	What it is
Build-up style	patient possession
Attacking width	stretch the back line
Vertical directness	runs behind the defense
Defensive aggression	pressing intensity

Data



Data Wrangling



Validated with error-checks on unmatched IDs — 0 observations removed.

Operationalization

Tactical Concept	Variable
Build-up style	Possession (%)
Attacking width	Corners
Vertical directness	Offsides
Defensive aggression	Fouls

Can You Predict the Results?

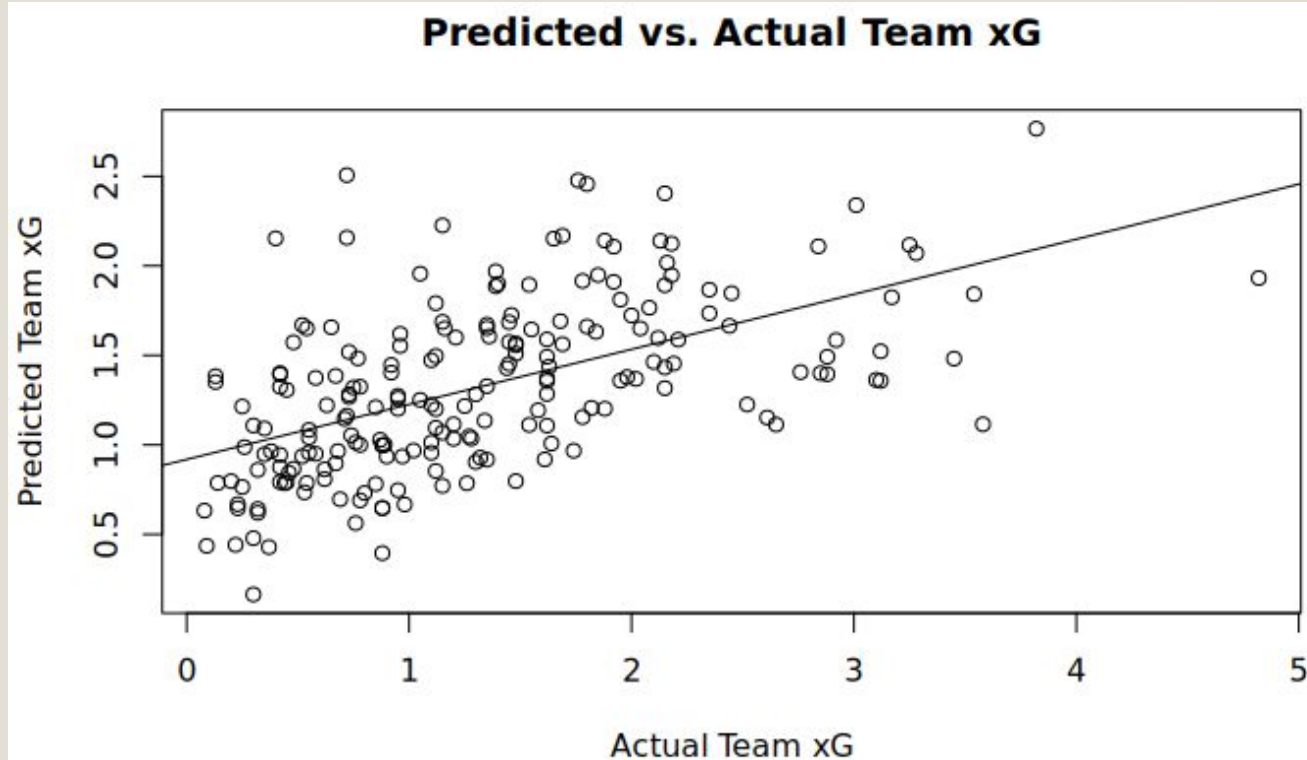
Question:

Which tactical factor do you think has the strongest relationship with expected goals (xG)?

-  Possession
-  Corners
-  Offsides
-  Fouls

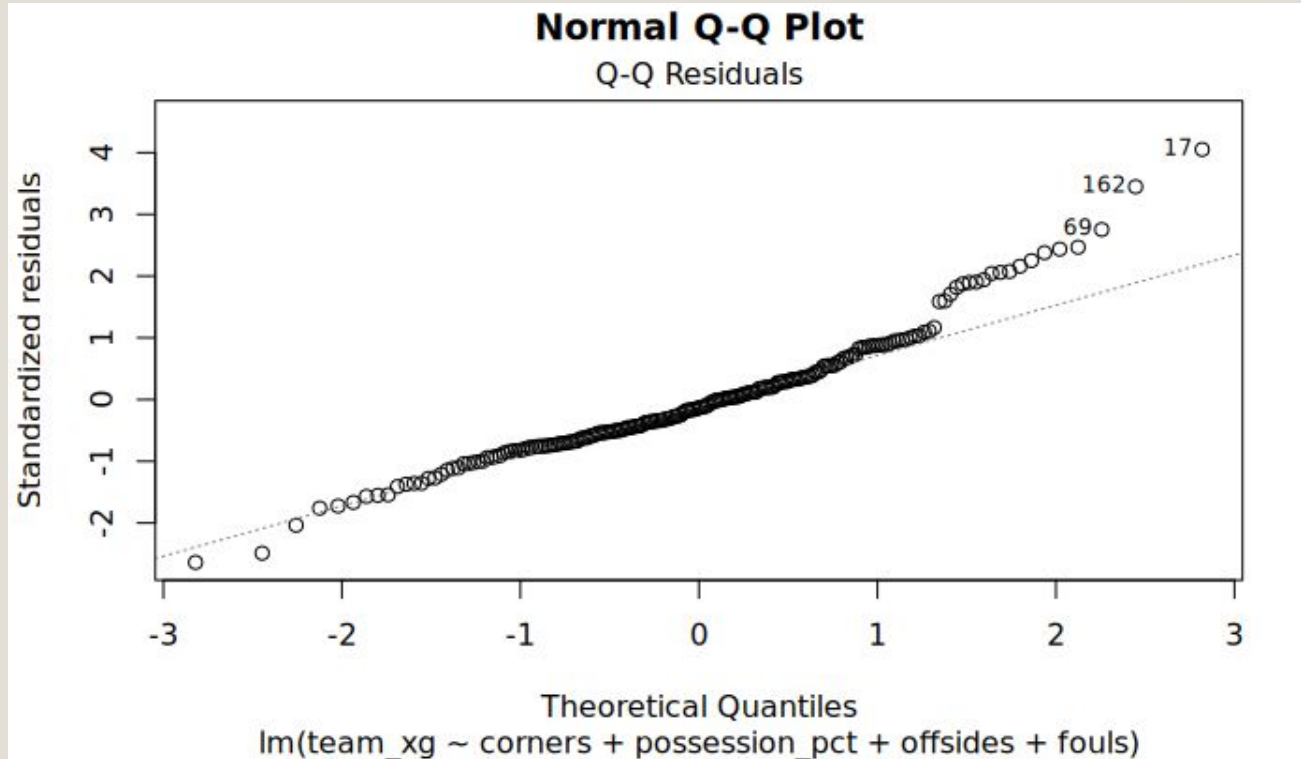
Data Visualizations

Comparing expected goals (xG) with estimated values from our linear regression model.



Data Visualizations Contd.

Evaluating the Q-Q plot for regression residuals. Residuals in the middle and towards the lower left tail indicate small deviations from the model. However, the right tail shows the model underestimates higher xG values



Model Specifications

Using expected goals (xG) as a dependent variable for offensive chance creation, leveraging possession percentage, corners created, offsides, and fouls as independent variables and proxies for game management and attacking intent/pressure

$$\text{Team_xG}_i = \beta_0 + \beta_1 \text{Possession}_i + \beta_2 \text{Corners}_i + \beta_3 \text{Offsides}_i + \beta_4 \text{Fouls}_i + \epsilon_i$$

Model Specifications Contd.

Regression results for specified model

Regression Results: Tactical Factors Associated with Team Expected Goals

Dependent variable:				
	(1)	Team Expected Goals (xG) (2)	(3)	(4)
Possession (%)	0.036*** (0.004)	0.027*** (0.005)	0.027*** (0.005)	0.024*** (0.005)
Corners Created		0.060*** (0.021)	0.060*** (0.021)	0.066*** (0.021)
offsides			0.008 (0.047)	0.012 (0.046)
Fouls				-0.037** (0.015)
Constant	-0.417* (0.213)	-0.266 (0.216)	-0.273 (0.221)	0.260 (0.301)
Observations	208	208	208	208
R2	0.257	0.285	0.285	0.307
Adjusted R2	0.253	0.278	0.274	0.294
Residual Std. Error	0.739 (df = 206)	0.727 (df = 205)	0.729 (df = 204)	0.719 (df = 203)
F Statistic	71.116*** (df = 1; 206)	40.829*** (df = 2; 205)	27.099*** (df = 3; 204)	22.531*** (df = 4; 203)

Note:

*p<0.1; **p<0.05; ***p<0.01

Model Assumptions

Multiple linear regression assumptions were evaluated using FIFA World Cup 2026 match data and diagnostic plots.

Linearity: Approximately linear relationship between predictors and xG

Independence (IID): Potential violation due to repeated team observations

Homoscedasticity: Residual variance approximately constant

Normality of Errors: The Q-Q plot indicated that residuals were approximately normally distributed, with minor deviations

Multicollinearity: Possession, corners, offsides and fouls may be related but represent distinct tactical dimensions.

Overall, the model provides descriptive evidence of tactical relationships with xG but should not be interpreted as causal due to factors such as opponent quality, player ability, and match context. The main limitation is potential dependence from repeated team observations.

Model Results and Interpretation

Regression Findings

Overall Model: Significant ($F = 22.53$, $p < .01$)

Model Fit: Explains **30.7%** of variation in xG ($R^2 = 0.307$)

	Positive/Negative	Effect	β	p value
Possession	Positive	Significant	0.024	$p < .01$
Corners	Positive	Significant	0.066	$p < .01$
Offsides	Positive	Not Significant	0.012	$p > .05$
Fouls	Negative	Significant	-0.037	$p < .05$

Overall Effect

Takeaway: Teams that maintained possession and consistently generated corner opportunities tended to produce higher expected goals. Fouls were negatively associated with xG, while offsides showed no statistically significant relationship with expected goals.

Tactical Behavior

What We Found

1

Possession

Positive, Significant ($P < 0.01$); Increase in Possession leads to an increase in xG

2

Corners

Positive, Significant ($P < 0.01$); Increase in Corners leads to an increase in xG

3

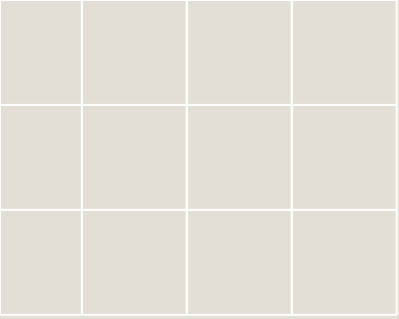
Fouls

Negative, Significant ($P < 0.05$); Increase in Fouls leads to a decrease in xG.

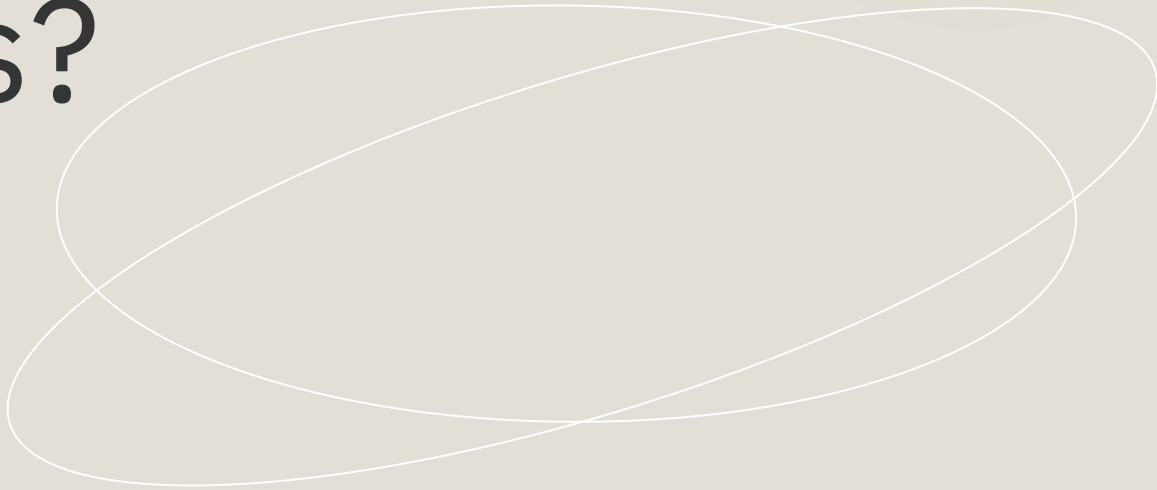
4

Offsides

Not Significant ($P \sim 0.865$)



Questions?



References

Data Sources:

Islam, MD Mominul. "FIFA World Cup 2026 Dataset - Live & Daily Updated Stats." Kaggle, version 1.0.0, 2026,
www.kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset.

Access Link:

- https://www.kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset?select=match_team_stats.csv
- <https://www.kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset/data?select=matches.csv>
- Data Prep & Access: Match statistics were merged and processed into match_team_stats_with_team_xg.csv to calculate team-level expected goals(xG) per match (N = 208)



Thank you